CHAPTER 8

COMPUTER NETWORKS

CHAPTER ESSENCE AT A GLANCE:
Computer networks connect autonomous computing systems across physical media and networking equipment to facilitate high-speed data transmission and resource sharing.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Network Classification by Geographic Span:** Networks scale from Personal Area Networks (PAN) spanning meters, to Local Area Networks (LAN) within buildings, Metropolitan Area Networks (MAN) across cities, to Wide Area Networks (WAN) spanning continents.

• **Physical Topologies and Redundancy:** Topological configurations define node wiring; Bus uses a central terminator backbone, Star routes through a central switch, Ring passes tokens sequentially, and Mesh connects every node directly.

• **Guided and Unguided Transmission Media:** Guided media encompass twisted pair cables, coaxial shielding, and total internal reflection in fiber optics, while unguided wireless media utilize radio waves, microwaves, and infrared signals.

• **Interconnection Hardware and Packet Relays:** Network interface cards provide physical attachment, hubs broadcast frames indiscriminately, switches filter frames by MAC address, and routers direct packets using IP routing tables.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Coverage Scale} = \text{PAN } (\sim 10 \text{ m}) < \text{LAN } (\sim 1 \text{ km}) < \text{MAN } (\sim 50 \text{ km}) < \text{WAN (Global)}$$

Geographic Span Tiers of Computer Networks

$$\text{Mesh Topology Cables} = \frac{n(n-1)}{2}, \quad \text{Ports per Node} = n - 1 \quad (n = \text{Node Count})$$

Full Mesh Topology Cable and Port Equation

$$\text{Guided Media} = \text{Twisted Pair (UTP / STP)} + \text{Coaxial Cable} + \text{Optical Fiber (TIR Light)}$$

Physical Guided Transmission Media Suite

$$\text{Hardware Tiers} = \frac{\text{Hub (Layer 1 Physical Broadcast)} \rightarrow \text{Switch (Layer 2 MAC Filter)}}{\text{Router (Layer 3 IP Routing)} \rightarrow \text{Gateway (Protocol Converter)}}$$

Network Interconnection Hardware Hierarchy

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** A network hub broadcasts packets to all connected ports causing collisions, whereas a network switch directs packets selectively to destination MAC addresses.

• **Alert:** Optical fiber cables transmit data as light pulses via total internal reflection, making them completely immune to electromagnetic interference.

• **Alert:** In Star topology, if an individual node cable fails only that node disconnects, but if the central hub or switch fails the entire network halts.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define a computer network. State any two major advantages of networking. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Definition: An interconnected collection of autonomous computers linked together via transmission media to facilitate data exchange and resource sharing. **[1 Mark]**

• Two Advantages: Hardware resource sharing (sharing expensive laser printers, storage arrays) and high-speed communication (instantaneous email and file transfers). **[1 Mark]**

2. Differentiate between Local Area Network (LAN) and Wide Area Network (WAN). [Score: 2 Marks]

Answer & Key Valuation Steps:

• Geographic Coverage: A LAN is restricted to a compact localized area like an office or school campus; a WAN spans extensive regional, national, or global boundaries. **[1 Mark]**

• Data Rate and Error Rate: LAN provides very high data transmission speeds (up to 1 Gbps) with negligible error rates; WAN operates at variable speeds with higher error rates due to long-distance public links. **[1 Mark]**

3. Explain Star topology and Bus topology with structural descriptions and their merits. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Star Topology: Every network host connects directly to a central hub or switch through point-to-point cables. Merits: Easy to install, simple fault detection, and failure of one cable does not disrupt other nodes. **[1.5 Marks]**

• Bus Topology: All workstations share a single linear central transmission backbone terminated at both ends. Merits: Minimal cable requirements and economical setup for small networks. **[1.5 Marks]**

4. Compare twisted pair cable, coaxial cable, and optical fiber cable in terms of bandwidth and EMI immunity. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Twisted Pair Cable: Moderate bandwidth, susceptible to electromagnetic interference (EMI); inexpensive and widely used in telephone and local office networks. **[1 Mark]**

• Coaxial Cable: High bandwidth, shielded against external electromagnetic noise; used in cable television systems. **[1 Mark]**

• Optical Fiber Cable: Immense bandwidth, transmits data as modulated light pulses; 100 percent immune to electromagnetic interference and ideal for long-distance telecommunications. **[1 Mark]**

5. Distinguish between a network hub and a network switch. Why is a switch called an intelligent hub? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Hub (Broadcast Device): Operates at physical layer; when data arrives at one port, it blindly broadcasts the electrical signal to all connected ports, causing network congestion and collisions. **[1 Mark]**

• Switch (Intelligent Filter): Operates at data link layer; inspects destination MAC addresses in packet headers and delivers data selectively to the intended destination port only. **[1 Mark]**

• Intelligence Justification: Because a switch builds an internal MAC address table dynamically and creates dedicated private communication paths, eliminating collisions, it is termed an intelligent hub. **[1 Mark]**

6. Explain the functions of a Router and a Gateway in internetwork communication. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Router Function: Operates at the network layer to interconnect different logical subnets, analyzing logical IP addresses to determine the optimal routing path for packet forwarding. **[1.5 Marks]**

• Gateway Function: An advanced protocol converter operating across all layers that connects networks with completely different architectures and transmission protocols, translating data formats seamlessly. **[1.5 Marks]**

7. A school has 4 separate buildings situated within 200 meters of each other. Suggest the most suitable network topology, transmission medium, and hardware devices to connect all computers. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Network Topology: Star topology inside each building connected to a central building via an extended Star backbone, ensuring fault isolation and simple maintenance. **[1 Mark]**

• Transmission Medium: Optical fiber cable for inter-building links to eliminate lightning surge risks and ground potential differences; Cat-6 UTP twisted pair cable for internal laboratory cabling. **[1 Mark]**

• Hardware Interconnection Devices: A managed network switch in each computer laboratory and a main distribution switch or router in the administrative server room. **[1 Mark]**

• Broadband Sharing Device: A broadband router with integrated firewall to connect the entire school network securely to the Internet service provider. **[1 Mark]**